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An integrated modeling approach to sustainable transport in forest biomass supply chains

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ABSTRACT

Sustainable forest biomass supply chains face increasing pressure to reduce environmental emissions while keeping costs low. Current research often underestimates the long-term impacts of adopting hybrid and electric truck technologies, as well as how these interact with government investments and fuel price uncertainties. This study addresses this gap by developing a digital twin–ready framework that integrates agent-based modeling and Geographic Information Systems to assess environmental and economic trade-offs in the forest biomass supply chain in Victoria, Australia, from 2025 to 2050. The framework compares diesel, hybrid, and battery-electric trucks across light, medium, and heavy classes under different supplier–tier configurations to measure emissions of CO_2, NO_x, SO_2, and PMc alongside transportation costs. Validation against empirical mass flow data for Level 1 suppliers across five scenarios showed average discrepancies of 0.5–3%, confirming the robustness of the simulation mode. Results reveal that electrification of heavy trucks yields the greatest sustainability benefits: up to 75% lower CO_2 and 70% lower PMc emissions compared to diesel, with only a 20% increase in transportation costs by 2050 under projected grid decarbonization. Hybrid trucks offer intermediate benefits, reducing CO_2 emissions by approximately 81% while maintaining cost parity with diesel. Light diesel trucks remain the least sustainable, emitting up to four times more NO_x and SO_2 per tonne-kilometre than electric alternatives. Social cost analysis highlights CO_2 and NO_x as the main externalities, exceeding 91% and 88% respectively by 2050 in diesel-heavy scenarios. The effect of technology is evident as electric trucks consistently outperform hybrids across all categories, with emissions reductions of 88% vs. 69% for light trucks, 81% vs. 50% for medium trucks, and 76% vs. 35% for heavy trucks, considering projected grid decarbonization of mobility, technology, and plans. The findings provide clear cost–emission trade-offs to inform fleet technology choices, supplier strategies, and policy investments for sustainable biomass logistics.

1. Introduction

Global demand for modern bioenergy is expected to increase substantially in the coming decades [1], with the IPCC projecting that bioenergy could contribute over 27% of the global energy mix by 2050 under 1.5 °C-aligned pathways [2]. In the European Union, forest biomass demand may exceed supply due to conversion inefficiencies and supply chain (SC) constraints [3]. Australia faces similar pressures under its legislated target to reduce emissions by 43% below 2005 levels by 2030 [4]. Although bioenergy currently contributes about 4% of national energy—compared with hydro (16%) and wind (12%)—its share could rise to 20% by the 2050 s [5], with solid and woody biomass accounting for 37% and 18% of renewable energy use, respectively [6]. Yet, forest biomass remains underutilized in Australia due to limited policy support, with residues often left to decompose or burned [7]. Addressing this gap requires an integrated assessment of biomass availability, harvesting practices, SC costs, and greenhouse gas (GHG) emissions across harvesting, storage, transport, and conversion processes [8].

Transportation is a central bottleneck in the sustainability of the forest biomass supply chain (BSC). Trucks account for only 4% of vehicles in Australia, but generate 38% of transport emissions [9], and freight demand is projected to double by 2050 [10]. While fleet electrification is increasingly recognized as essential for logistics

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decarbonization, research has largely focused on passenger transport, with limited attention to heavy-duty biomass logistics [10–12]. As a result, diesel-dominated biomass transport continues to emit substantial carbon dioxide (CO_2), nitrogen oxides (NO_x), sulfur dioxide (SO_2), and particulate matter (PMc) [13] and existing studies often underestimate the long-term cost and emission implications of hybrid and electric trucks under evolving fuel prices and policy incentives [14]. These limitations highlight the need for modelling approaches that can capture heterogeneous vehicle technologies, spatially distributed suppliers, and dynamic operational decisions.

Supply Chain Digital Twins (SCDTs) offer a promising pathway to address these challenges by enabling data-driven replication of real-world SC structures and behaviours for scenario analysis and decision support [15]. Their application in bioenergy and waste-to-energy systems has demonstrated potential for improving efficiency, emissions performance, and conversion processes [16], particularly when integrated with predictive analytics and machine learning [17]. However, large-scale adoption remains constrained by data availability, feedstock variability, and computational demands [18]. While Industry 4.0 technologies have enabled digitally interconnected SCs, evidence suggests that isolated technological interventions alone are insufficient to achieve meaningful GHG reductions [19], especially in transportation-intensive biomass SCs where spatial and behavioural dynamics play a dominant role [8,20].

To address these methodological gaps, this study positions an integrated Agent-Based Modeling (ABM) and Geographic Information System (GIS) framework within the SCDT paradigm. This approach enables explicit representation of spatial heterogeneity, agent-level decision-making, and spatiotemporal variability in emissions and costs across vehicle technologies and supplier tiers; limitations that are not adequately addressed by optimisation-only or static lifecycle approaches [21,22]. By supporting dynamic "what-if" analysis and forward-looking optimisation, the proposed ABM–GIS–DT framework aligns operational decision-making with organizational Sustainable Development Goals (SDGs) [23], advancing SDGs 7 (Affordable and Clean Energy), 9 (Industry, Innovation, and Infrastructure), and 13 (Climate Action) through improved quantification of freight electrification impacts and resilient forest biomass SC design [24–26]. This methodological positioning provides a scalable decision-support foundation for achieving net-zero BSC under increasing regulatory and policy pressure [27,28].

2. Literature review

Currently, there is limited information on large-scale electrification of freight transport in Australia, particularly for heavy-duty trucks, rail, and shipping, compared to China, India, and Europe [29]. Most of the literature on BSC transportation assumes diesel-powered trucks as the default for biomass transportation and identifies them as a major contributor to the feedstock supply cost and associated GHG emissions [30].

The deployment of advanced vehicle technologies, including fuel cell, hybrid, and fully electric powertrains, for biomass feedstock transport represents a largely unexplored strategy for lowering biofuel production costs, reducing GHG emissions, and mitigating associated health impacts [30]. Life Cycle Assessments (LCAs) have become a key tool for evaluating transport sustainability, as they capture full life cycle impacts beyond tailpipe emissions [31]. Existing studies show substantial heterogeneity in scope, methodology, and assumptions, limiting cross-study comparability [32]. For example, some analyses adopt a well-to-wheel perspective, focusing on fuel production and vehicle operation emissions [33], while others take a cradle-to-grave approach, covering raw material extraction, vehicle use, and end-of-life treatment [34]. Inconsistencies also arise from variations in vehicle classification, mission types, annual mileage, and lifetime assumptions [35,36].

Most vehicle-focused LCA studies have concentrated on alternative fuels and powertrains [37]. While some reviews consider broader transport modes [38], there remains a lack of integrated analyses specifically comparing diesel, hybrid, and battery-electric trucks in heavy-duty applications. Consequently, the existing literature provides limited guidance on the combined environmental and economic trade-offs of fleet electrification strategies, particularly under realistic SC scenarios. This gap motivates the present study, which integrates GIS and ABM within a DT framework to evaluate multiple truck types across tiered forest BSCs, addressing both emissions and cost implications from 2025 to 2050.

2.1. Forest BSC transportation and logistics

The literature on forest BSC logistics has primarily focused on economic optimization, route planning, and cost-effective harvesting and transportation methods using GIS, simulation, and, to a lesser extent, multi-criteria decision-making (MCDM). According to Table 1, early GIS-based studies focused on strategic-level decision-making, such as assessing biomass availability, delivery costs, and optimal harvesting methods, yet they largely overlooked environmental impacts, particularly emissions. Simulation studies, including discrete-event and dynamic simulations, incorporated environmental considerations, primarily GHG emissions, SO_2, and energy consumption, while also evaluating transportation costs.

Some studies extended this analysis to multimodal transport, including trucks, trains, barges, and pipelines, to evaluate cost-efficiency and emission trade-offs [39]. Digital twin applications in the biomass sector [40] have started to emerge, highlighting operational efficiency and real-time monitoring of hybrid and electric trucks; however, most existing studies remain limited in scope, often focusing on energy production rather than integrating transportation sustainability with predictive, scenario-based frameworks.

Despite the advances in GIS, simulation, and DT applications, significant gaps remain in the literature. First, few studies provide a comprehensive integration of economic and environmental performance metrics across multiple truck types, including traditional diesel, hybrid, and electric vehicles, particularly in predictive future mobility scenarios toward 2050.

Second, the majority of prior work emphasizes strategic-level planning with limited tactical or operational insights, leaving a gap in scenario-based evaluation of fleet composition, supplier allocation, and route optimization under varying government investment and fuel price conditions. Third, existing studies rarely incorporate the combined strengths of ABM, GIS, and DT technologies to simulate realistic, dynamic SC interactions while capturing spatial variability and policy impacts.

Finally, there is a lack of research that extends findings beyond the biomass sector or Victoria, Australia, to broader SCs and global contexts, limiting the applicability of lessons learned for emerging sustainable transport and mobility technologies. This study addresses these gaps by integrating DT simulations with ABM and GIS, assessing hybrid and electric truck adoption under multiple investment and fuel price scenarios, and evaluating both environmental emissions (CO_2, NO_x, SO_2, PMc) and transportation costs for different types of trucks. The approach provides a holistic, forward-looking framework for sustainable mobility in biomass and other SCs.

2.2. Supply chain digital twin framework

Recent studies have examined the intersection of SC management and digitalization through concepts such as smart green SCs, SC 4.0, and I4.0-enabled sustainable SCs [51,52]. However, these frameworks remain fragmented, lacking an integrated approach that links digital intelligence with sustainability performance, particularly for monitoring and mitigating transport-related emissions and costs.

DT technology offers a promising pathway to address this gap by creating virtual replicas of logistics systems for real-time monitoring,

Table 1
Summary of studies on BSC transportation using GIS and computer-based simulation models.

Author	Method type	Objective of study	Decision levels Type of transport	Sustainability	Type of emission
GIS					
[41]	GIS	To assess the availability of woody biomass and the cost of its delivery.	✓ Strategic • Truck • Truck	• Economic	×
[42]	GIS	To analyze economic factors for evaluating the sustainability and viability of establishing a switchgrass biorefinery.	✓ Strategic ✓ Tactical • Truck • Truck	• Economic • Environmental	×
[43]	GIS	To develop a cost model for identifying the most economical harvesting method.	✓ Strategic ✓ Operational • Truck	• Economic	×
[44]	GIS	To establish a method for determining the optimal utilization of forest biomass.	✓ Strategic • Truck	• Economic	×
[45]	GIS and MCDM	To examine the logistics of converting biomass to electricity.	✓ Strategic ✓ Tactical • Truck	• Economic	• GHG
Computer Simulation					
[46]	Simulation model	To assess the harvesting and transportation systems for sugarcane and mills	✓ Strategic • Truck	• Economic	×
[47]	Discrete-event simulation (DES)	To optimize truck scheduling within the biomass transportation network.	✓ Strategic • Truck	• Economic	×
[48]	(DES)	To analyze GHG emissions and delivery costs within the woody residues BSC.	✓ Strategic • Truck	• Economic • Environmental	• GHG
[49]	(DES)	To assess the effect of different chipper and transportation types on the energy consumption and costs.	✓ Strategic ✓ Tactical • Truck	• Economic	×
[8]	Dynamics simulation	To assess the impact of delivery and production technologies on the environmental emissions.	✓ Strategic ✓ Tactical • Truck • Train • Truck	• Economic • Environmental	• GHG
	Dynamics simulation	To analyze the trends in transportation costs and GHG emissions.	✓ Strategic ✓ Tactical • Truck • Train	• Economic • Environmental	• GHG
	Dynamics simulation and life cycle assessment	To forecast the water-energy nexus and evaluate the environmental emissions of the BSC over 50 years.	✓ Strategic ✓ Tactical • Truck • Train • Pipeline	• Economic • Environmental	• GHG • SO_x
	Dynamics simulation	To examine the transportation costs and environmental impacts of the BSC by evaluating three different transportation modes.	✓ Strategic ✓ Tactical • Truck • Train • Barge	• Economic • Environmental	• GHG
	Dynamics simulation	To assess particulate air pollutants and cost per mass flow associated with maritime transportation in the global BSC.	✓ Strategic ✓ Tactical ✓ Operational • Barge	• Economic • Environmental	• GHG • SO_2 • NO_x • PM_c
DT in BSC					
[40]	Digital Twin	To demonstrate how DT technology enables energy companies to optimize forest BSC by simulating operational scenarios to assess hybrid and electric truck transportation in terms of efficiency, emissions, and resource management.	• Strategic • Hybrid Truck • Electric Truck	• Economic	×
[50]	Digital Twin	To focus on the role of DT technology in bioenergy production, emphasizing real-time monitoring, process simulation, and hybrid energy systems rather than transportation logistics. However, the paper does not address the optimization of sustainable truck transportation in forest BSCs.	• Strategic	• Economic	×
DT Ready Simulation and GIS					
Current Study	DT /Integration of GIS and ABM	To analyse the mobility of Sustainable BSC by 2050	✓ Strategic ✓ Tactical ✓ Hybrid ✓ Electric ✓ Traditional • Light • Medium • Heavy	• Economic • Environmental	• CO_2 • SO_2 • NO_x • PM_c

scenario testing, and optimization. When combined with GIS, IoT, and ABM, DTs enable predictive maintenance, route optimization, and emission reduction, demonstrating up to 80% improvement in operational efficiency and sustainability [26]. In Australia, where policies such as the National Electric Vehicle Strategy promote low-emission mobility, DT-enabled modeling can support evidence-based planning for fleet transition, infrastructure investment, and policy evaluation [53].

Moreover, DT applications have been successfully adopted in manufacturing and agriculture for process optimization, circular economy integration, and logistics efficiency enhancement [26]. These studies highlight DTs' ability to improve inventory control, fleet management, and energy forecasting. However, their application to natural resource-based SCs, especially in forestry and biomass logistics, remains limited. Building on these insights, this study developed a framework illustrated in Fig. 1 that integrates the technological and methodological dimensions of I4.0 to support the transition toward a net-zero BSC by 2050, directly aligning with the key contributions of this study in the next section. The core of the framework reflects the proposed agent-based and GIS-integrated simulation model, which serves as the foundation for evaluating environmental and economic trade-offs in multimodal transportation systems. Surrounding this core, the inclusion of DT, modeling and simulation, optimization, and decision-making components demonstrates how our research contributes to building a dynamic, data-driven decision-support system for sustainable SC management. Furthermore, the outer layers of the framework capture emerging I4.0 technologies such as IoT, AI, cloud computing, and cyber-physical systems, positioning them as potential future extensions to enhance real-time data acquisition, predictive analytics, and adaptive optimization. These technologies, while beyond the current study's scope, represent significant avenues for future exploration not only within biomass logistics but also across broader industrial SCs aiming to achieve net-zero transportation systems by 2050.

2.3. Contribution of the study

This study makes several key contributions to advancing sustainable transportation research in forest BSCs, particularly in the context of achieving Australia's and global net-zero targets by 2050. First, it develops an integrated simulation framework that combines ABM with GIS to capture both the spatial heterogeneity and behavioral dynamics of forest biomass logistics. Unlike prior work, which has typically focused on cost or emissions in isolation, this framework evaluates the environmental and economic trade-offs of multiple truck types, including light, medium, heavy, hybrid, and electric, across different supplier tiers and transportation scenarios. By quantifying emissions (CO_2, NO_x, SO_2, PMc) alongside their associated costs, the model identifies critical emission hotspots and cost drivers, thereby providing stakeholders with actionable insights for designing optimized low-emission strategies.

Second, the study aligns its simulation design with the concept of DTs, offering a step toward real-time, data-driven decision-making for SC sustainability. By demonstrating how ABM and GIS can be integrated into a DT framework, the research not only supports scenario-based planning but also establishes a foundation for future incorporation of IoT-enabled live data streams, predictive maintenance, and real-time routing optimization. This forward-looking perspective facilitates the assessment of transition pathways from current diesel fleets to hybrid and electric alternatives under varying technological, market, and

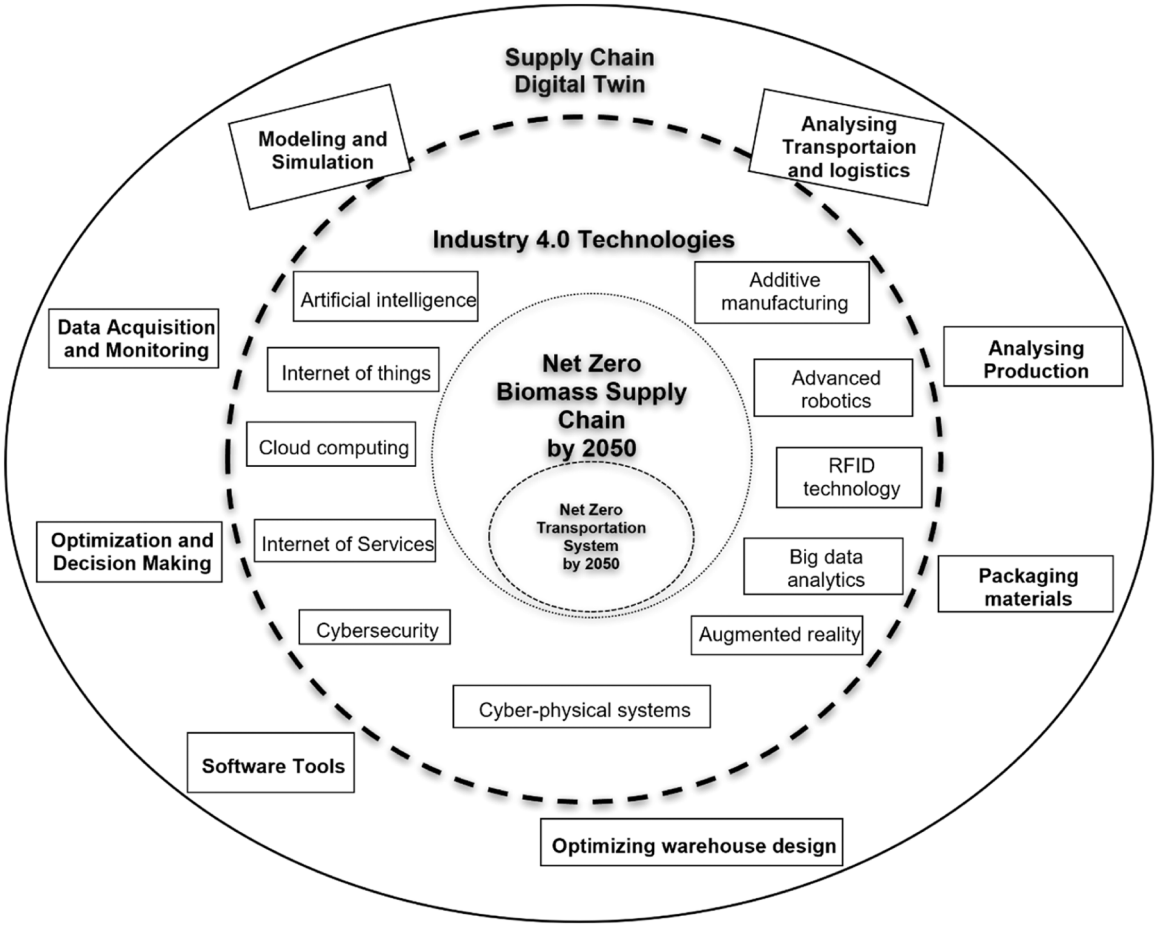


Fig. 1. Conceptual framework for achieving a Net Zero BSC by 2050 (created by authors).

policy conditions, providing an evidence base for long-term fleet electrification and infrastructure planning.

Third, the study advances the methodological frontier by bridging operational logistics with environmental modeling through a sensitivity analysis approach. This enables policymakers and industry stakeholders to evaluate trade-offs across cost efficiency, emission reductions, and policy compliance, supporting balanced and sustainable decision-making. The integration of digital twin-enabled modeling with scenario analysis extends beyond tactical operations to strategic planning, making the findings relevant at regional, national, and global scales.

Australia currently ranks sixth globally in the Renewable Energy Country Attractiveness Index, reflecting its strong potential for renewable energy and broader green economy investments. This momentum is driven by increased government funding, abundant natural resources, and the nation's strategic capacity to contribute solutions to global SCs. Complementing existing electric and hybrid vehicle initiatives, the Australian Government is further promoting investment and innovation across the electric vehicle SC. This research aligns with Australia's RACE for 2030 Cooperative Research Centre goals, addressing energy productivity, decarbonization, and I4.0 innovation in SCs [54]. By analysing sustainable biomass mobility, the study supports the development of scalable digital innovations that can guide the transition toward hybrid and electric truck adoption by 2050.

3. Materials and methods

3.1. Case study and data collection

Australia has significant woody biomass resources, with approximately 21 million cubic metres of hardwood and softwood logs used annually for lumber, panels, and paper products, supporting both domestic and international markets [55]. The cultivation of durable biomass crops offers additional environmental benefits, including carbon sequestration, mitigation of dryland salinity, biodiversity enhancement, and reduced wind erosion.

In Victoria, 11 biomass boilers are currently operational, utilizing woodchips and straw pellets for energy generation in the form of steam, electricity, and hot water. Biomass equipment manufacturers advocate woodchips as a biofuel, with larger boilers accommodating diverse feedstock sizes and moisture levels, while smaller units require consistent specifications. Regional suppliers provide cost-effective options such as wet or recycled woodchips. Fig. 2 indicates plantation areas capable of supplying between 10,000 and 456,000 tonnes across five major regions, highlighting the state's substantial biomass availability for sustainable energy applications [55].

Fig. 3 illustrates the existing woody BSC in Victoria, highlighting Tier 1 and Tier 2 suppliers and eight biorefineries producing heat, steam, and electricity for agricultural and institutional applications. Tier 1 suppliers provide diverse feedstocks, including kiln-dried, chipped, and low-moisture wood, with costs ranging from $40 to $100 per tonne, while Tier 2 suppliers process and deliver woodchips to biorefineries. Biorefineries integrate these inputs to reduce fossil fuel use, achieve cost savings, and, in some cases, produce biomass on-site to close the loop.

Transportation flows from suppliers to biorefineries are mapped across Victoria's road network using GIS [57], reflecting recent infrastructure investments exceeding $3 billion to improve freight efficiency. Road trains in the region can operate at up to 85.5 tonnes, further facilitating large-scale biomass transport.

3.2. Model development description

The developed GIS–ABM model was implemented in the AnyLogic platform to analyze transportation dynamics and environmental emissions in the woody BSC of Victoria, Australia. The model integrates spatial precision from GIS data with agent autonomy to represent suppliers, trucks, and biorefineries. Each agent operates within a structured set of states and activities. Trucks alternate between idle, loading, travelling, unloading, and return-to-depot states. Suppliers manage inventory and dispatch schedules, while biorefineries process incoming biomass according to their operational capacities. Agent movement and interactions are governed by AnyLogic's built-in geospatial functions, which simulate realistic road-based travel times and positional updates across the modeled geographic landscape. The developed model integrates a Java-based programming layer within the AnyLogic

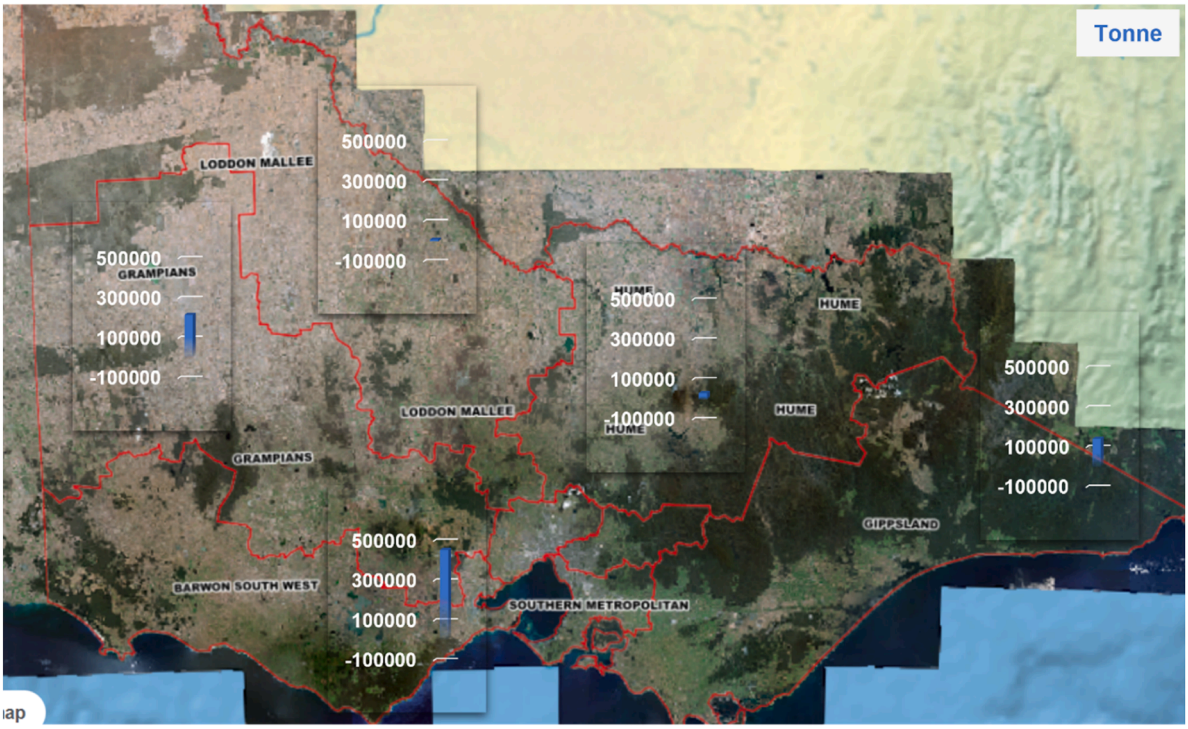


Fig. 2. Digital Twin VIC data [56] for forestry plantations [55].

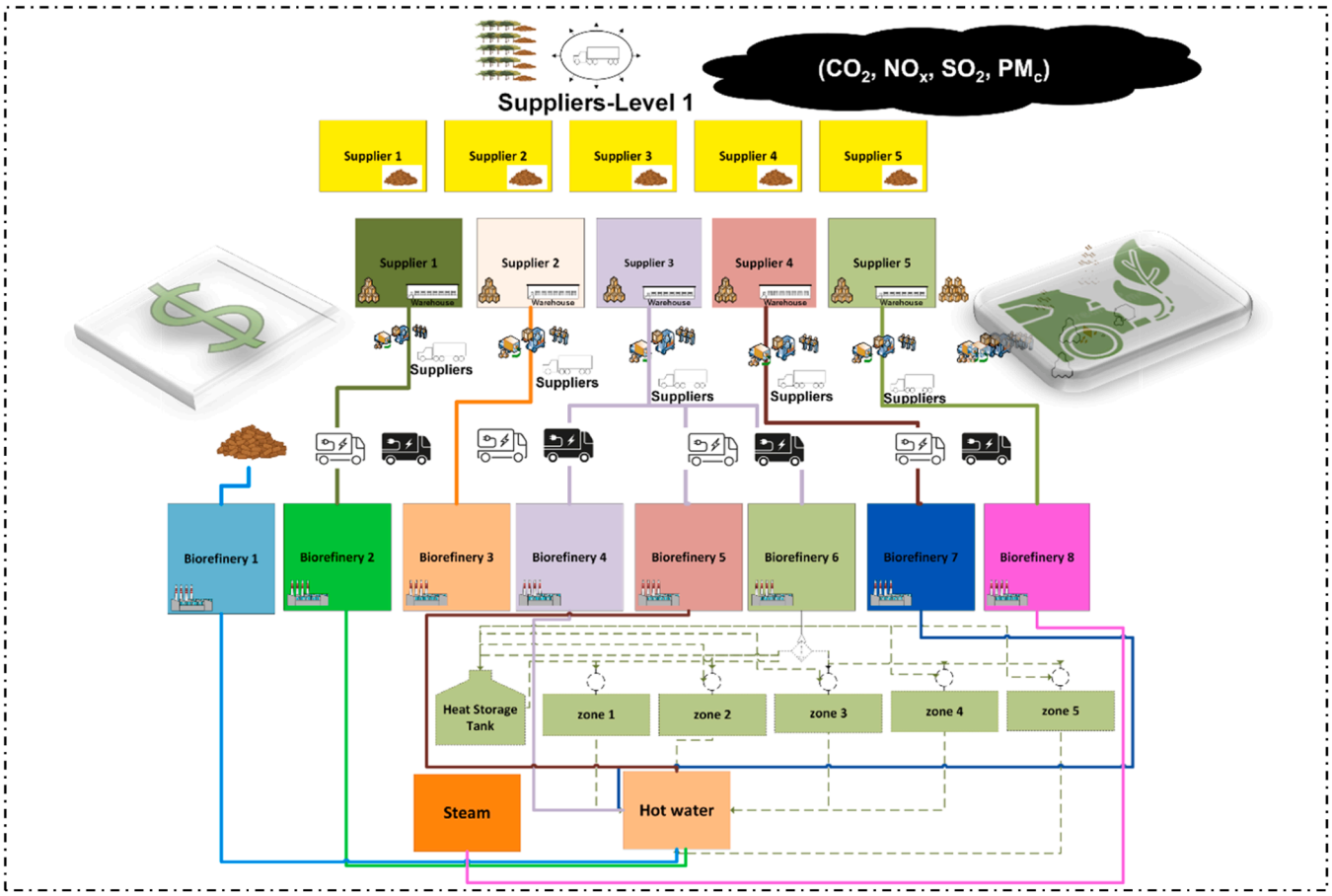


Fig. 3. Woody BSC Design in VIC state [57].

environment, enabling dynamic spatial behaviors and realistic agent interactions over the GIS map. These custom-coded functions manage agents' movements, spatial positioning, and interaction with geographic elements, providing fine-grained control over transportation flows and logistics operations in the BSC system.

Fig. 4 presents the DT framework of the integrated GIS-enabled agent-based and system dynamics model developed in this study. The simulation process begins with order generation at the biorefinery, where demand-driven orders are created and assigned to available suppliers based on predefined capacity, inventory, and spatial constraints. These orders trigger agent-level decision-making processes, including supplier selection, truck dispatching, and route assignment, which are dynamically informed by the GIS layer.

Truck agents execute a sequence of activities, order acceptance, waiting, loading, transportation, and delivery, while explicitly accounting for operational delays such as loading time, traffic conditions, and unloading processes. The GIS module continuously provides spatial information, including distances, routing, and network conditions, enabling a realistic representation of transportation dynamics.

At the system level, the system dynamics component aggregates agent-level outcomes to capture feedback effects on inventory levels, order fulfilment rates, transportation performance, and emissions. All model components are integrated and executed within the AnyLogic environment, where the root model coordinates agent interactions, simulation control, and data exchange between the GIS, agent-based, and system dynamics layers. This structured integration enables a transparent, end-to-end representation of the SCDT, clearly linking model flow, agent activities, decision logic, and the overall simulation process.

Decision-making in the system follows explicit rules aligned with real-world logistics logic. Truck dispatch is prioritized by order urgency, vehicle availability, and the shortest travel distance, while suppliers allocate resources based on inventory levels and storage constraints. The event calendar manages recurring operations, daily order generation, transportation dispatch, and biomass processing, ensuring temporal consistency between agent activities. All key model inputs, including biomass supply volumes, road distances, cost factors, and emission coefficients, were drawn from reliable datasets such as Sustainability Victoria [58], database reports [59,60], and prior empirical research [61]. The model outputs include transportation costs, pollutant emissions (CO_2, NO_x, SO_2, PMc), and system-level performance metrics, aggregated at daily, monthly, and annual levels for operational and strategic decision support.

To reflect real-world uncertainty in biomass logistics, stochastic distributions were applied to key variables such as biomass availability, delivery volume, and travel time. The residual mass flow for each supplier was analyzed to determine the most suitable probability distribution, ensuring statistical consistency and realistic variability in SC performance. As shown in Appendix Fig. A1, the best-fit distributions for the five suppliers were identified as gamma, lognormal, and normal, with p-values above 0.6 indicating good model fit. These distributions capture inherent fluctuations in mass flow caused by supply inconsistency and transport delays.

A one-month warm-up period was implemented to allow the system to reach steady-state conditions before collecting performance metrics, following established guidance for discrete-event and agent-based simulations [62,63]. Random-number seeding ensures reproducibility across simulation replications, and the number of replications was determined to achieve statistically reliable estimates, such that confidence intervals for key outputs remained within acceptable bounds,

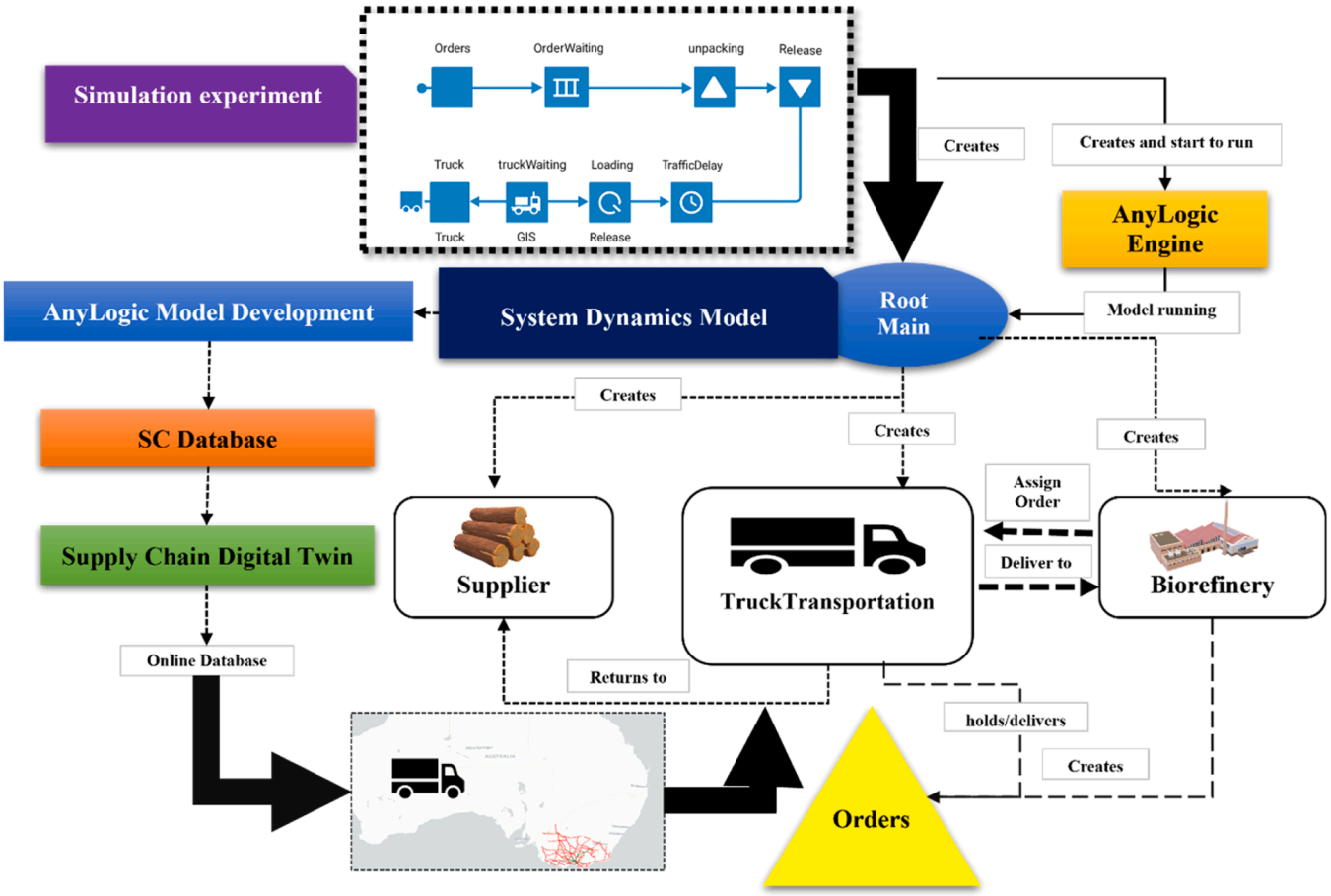


Fig. 4. DT Framework of the integrated GIS and ABM Model.

consistent with standard simulation practice [64]. The model operates at a daily temporal resolution within a GIS-based spatial layer of Victoria's transport network, enabling accurate route optimization, visualization of material flows, and assessment of the long-term impacts of fleet electrification on cost and emissions.

The chosen abstraction level achieves a balance between modeling fidelity and computational efficiency. Core processes, order allocation, transportation scheduling, and emissions accounting, are explicitly represented, while unnecessary micro-level operations are omitted to maintain transparency and replicability. The mathematical framework incorporates five categories of equations to compute production, transportation, storage, and emissions costs, including CO_2, SO_2, NO_x, and PMc. Three transport modes, including light, medium, and heavy container trucks, are evaluated, with detailed indices and variable definitions provided in Table 2. Unit production costs for growing, harvesting, collection, and chipping are set at $15, $50, and $25 per tonne [65]. Screenshots and diagrams from AnyLogic illustrate key agent states, data flows, and simulation outputs, ensuring full transparency and enabling replication by future researchers. Table 2 details the core geospatial functions and parameters applied in the model. Table A1 in the Appendix lists Java-based agents that were used to operationalize the GIS-enabled simulation. Fig. 5 illustrates the transportation distances for woody residues from level 1 suppliers to level 2 suppliers, and subsequently from level 2 suppliers to biorefineries. Additionally, it provides a GIS-based visualization of Victoria's existing roadway network, sourced from [57].

Table 3 outlines five distinct scenarios for the Woody BSC transportation system, differentiated by the type and capacity of trucks-light, medium, and heavy-assigned to various supplier levels. This predictive scenario analysis was built into the agent-based simulation model to evaluate how varying truck capacities affect environmental emissions and transportation costs across the SC. Each scenario reflects a strategic allocation of vehicle types to supplier locations based on operational feasibility, distance, terrain, and load requirements. For instance, Scenario 1 predominantly uses light trucks except for Supplier (S) 1 at level (L) 1, which employs heavy trucks. This setup may reflect last-mile or short-distance delivery patterns, allowing the model to evaluate emissions and costs under lightweight, more flexible operations.

In contrast, Scenarios 2 through 5 introduce different transportation configurations to assess their comparative impacts. Scenario 2 emphasizes medium and heavy trucks, indicating a shift toward higher-capacity, long-haul strategies. Scenario 3 distributes truck types more diversely among suppliers to simulate a hybrid model. Meanwhile, Scenario 4 exclusively employs heavy trucks for all suppliers, representing a high-capacity, potentially cost-effective but emission-intensive setup. Lastly, Scenario 5 applies medium trucks only, providing a middle-ground strategy. This multi-scenario framework allows for predictive modeling of emissions (CO_2, NO_x, SO_2, PM) and cost trade-offs, informing the selection of optimal transportation configurations in terms of environmental and economic performance in the BSC.

3.2.1. *Model verification, validation, and credibility*

The developed GIS–ABM model was rigorously verified and validated to ensure reliability and reproducibility. Verification procedures included structured code walkthroughs, animation-based checks of agent movements and interactions, extreme-value testing, and trace-based debugging to ensure that model logic and event handling were correctly implemented. The model outputs were validated against empirical mass flow data for Level 1 suppliers across five simulation scenarios. Fig. 6 presents a comprehensive visualisation of the

Table 2
Definitions of indices, parameters, and variables in the simulation model.

Index	Definition
m = (1,…,M)	Index for Primary Raw Material Sources
n = (1,…,N)	Index for Intermediate Collection Points
r = (1,…,R)	Index for Biorefinery Facilities
p = (1,…,P)	Index for Biomass Product Types
v = (1,…,V)	Index for Transport Modes
c = (1,…,C)	Index for Transport Capacities
Parameters	**Definition**
$\mathbf{RM_m}$	Available woody biomass at source m (tonnes)
$\mathbf{Q^p_{mn}}$	Quantity of biomass product p transported from raw source m to collection point n (tonnes)
$\mathbf{Q^p_{nr}}$	Quantity of biomass product p transported from collection point n to biorefinery r (tonnes)
$\mathbf{EM^{RM}_{mnvc}}$	Emissions intensity (GHG (CO_2-eq), SO_2, PM_c, or NO_x) for transportation from m to n using mode v and capacity c (g/tkm)
$\mathbf{PR^p_r}$	Quantity of biomass product p produced at biorefinery r (tonnes)
$\mathbf{EM^p_{nrvc}}$	Emissions intensity (GHG (CO_2-eq), SO_2, PM_c, or NO_x) for transportation from n to r using mode v and capacity c (g/tkm)
$\mathbf{D^p_{mn}}$	Transportation distance from source m to collection point n for product p (km)
$\mathbf{D^p_{nr}}$	Transportation distance from collection point n to biorefinery r for product p (km)
$\mathbf{CP^p_r}$	Unit production cost of biomass product p at biorefinery r ($/tonne)
$\mathbf{TC^{RM}_{mn}}$	Transportation cost from m to n for raw material ($/tonne/km)
$\mathbf{TC^p_{nr}}$	Transportation cost from n to r for product p ($/tonne/km)
$\mathbf{SC_p}$	Storage cost per tonne of product p at biorefinery ($/tonne)
$\mathbf{ST^p_r}$	Quantity of biomass product p stored at biorefinery r (tonnes)
Variables	**Definition**
Cost_RM	Cost of biomass raw material acquisition ($/tonne)
Cost_Em	Cost of emissions ($/tonne)
Cost_TotalRM	Total cost of raw biomass ($/tonne)
Cost_Trans	Total transportation cost ($/tonne)
Cost_Prod	Total production cost at biorefineries ($/tonne)
Cost_EmTrans	Emission-related transportation cost ($/tonne)
Cost_Storage	Total storage cost ($)
$\boldsymbol{EMTotal^p_{mnvc}}$	Total emissions from transporting product p from source m to collection point n via mode v and capacity c (tonnes CO_2-eq/year)
$\boldsymbol{EMtotal^p_{nrvc}}$	Total emissions from transporting product p from collection point n to biorefinery r via mode v and capacity c (tonnes CO_2-eq/year)
Equations	**Definition**
(1)	Calculate emissions from transportation between m and n across transport modes and capacities
	$EMTotal^p_{mnvc} = Q^p_{mn} * EM^{RM}_{mnvc} * D^p_{mn} \forall_{m,n,p,v,c}$
(2)	Calculate emissions from transportation between n and r transport modes and capacities
	$EMtotal^p_{nrvc} = Q^p_{nr} * EM^p_{nrvc} * D^p_{nr} \forall_{n,r,p,v,c}$
(3)	Compute total transportation cost from m → n and n → r
	$Cost_Trans = ((\sum_m^M \sum_n^N \sum_p^P Q^p_{mn} * TC^{RM}_{mn}) + (\sum_n^N \sum_r^R \sum_p^P Q^p_{nr} * TC^p_{nr}))$
(4)	Calculate total cost of biomass procurement, including raw material harvesting at source m
	$Cost_TotalRM = (\sum_m^M RM_m * Cost_RM)$
(5)	Compute production costs at each biorefinery r
	$Cost_Prod = ((\sum_r^R \sum_p^P PR^p_r * CP^p_r)$
(6)	Calculate total emission costs by converting transport emissions into monetary values
	$Cost_EmTrans = ((\sum_m^M \sum_n^N \sum_p^P \sum_v^V \sum_c^C EMTotal^p_{mnvc}) + (\sum_n^N \sum_r^R \sum_p^P \sum_v^V \sum_c^C EMtotal^p_{nrvc})) * Cost_Em))$
(7)	Estimate storage costs based on material held at biorefinery r
	$Cost_Storage = (\sum_r^R \sum_p^P ST^p_r * SC_p)$

simulation model validation. The top panel displays scatter points of simulation versus actual mass flow for all scenarios along with the actual mass flow as a dashed reference line, while the bottom panel illustrates the residuals (differences) as bars to quickly identify over- or underestimation. Across the scenarios, the average discrepancies between simulated and actual mass flows ranged from approximately 0.5% to 3%, indicating strong alignment and robust predictive capability. Sensitivity analyses were conducted to evaluate the influence of key parameters such as supplier delivery volumes, transport distances, and vehicle assignment rules, confirming that the model consistently captures system dynamics under varying operational conditions. These procedures collectively demonstrate the accuracy and reliability of the model for strategic and operational analysis in BSC simulations.

The credibility of the developed GIS–ABM framework is supported using authoritative data sources and prior research. Model parameters, input distributions, and assumptions are informed by published studies, government reports, and expert datasets (e.g., Sustainability Victoria [58], database reports [59,60], and prior empirical research [61]. By grounding the model in established literature and validated empirical data, we ensure that the simulation outputs are consistent with real-world behaviors of woody BSCs. This approach provides confidence in the robustness and reliability of the model, even in the absence of direct stakeholder engagement, and facilitates its use for predictive scenario analyses and strategic decision-making.

4. Results and discussion

4.1. Environmental emission analysis

Fig. 7 summarizes the environmental emissions (CO_2, NO_x, SO_2, PMc) across five transport scenarios, highlighting the influence of truck type and capacity on SC performance. Scenario 1, which predominantly employs light trucks, consistently produced the highest emissions across all pollutants. For example, CO_2 emissions at Supplier 5 Level 2 reached 130.06 million g, reflecting the inefficiency of small vehicles for long distances and heavy loads, which necessitate multiple trips. Transitioning to mixed heavy and medium trucks in Scenario 2 markedly reduced emissions; Supplier 5 Level 2 CO_2 dropped by 52.7% relative to Scenario 1. Scenarios 3 and 5, using mixed or medium trucks, demonstrated intermediate outcomes, while Scenario 4, using only heavy

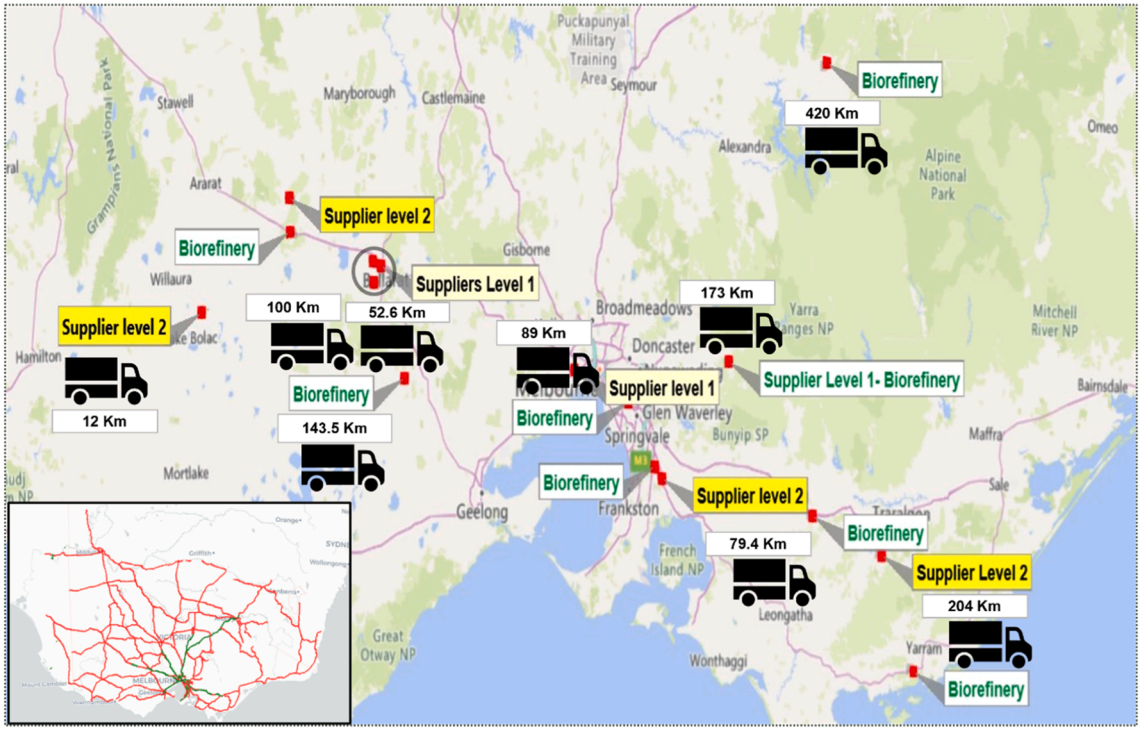


Fig. 5. Transportation distances between suppliers along with a GIS-based map of Digital Twin VIC's road network [57].

trucks, achieved the lowest CO_2 and NO_x levels across most suppliers.

Comparable trends were observed for NO_x, SO_2, and PMc (Fig. 7b–d), confirming that vehicle capacity and fuel efficiency dominate emission outcomes across pollutant types. Heavy-duty trucks consistently reduced emissions by minimizing trip frequency and maximizing load efficiency, whereas extensive reliance on light trucks amplified cumulative emissions. Medium trucks demonstrated balanced performance, offering flexibility while moderating environmental impacts, particularly for routes or suppliers where heavy truck deployment is constrained.

To enable comparison across emission types and scenarios, emissions were normalized relative to annual totals for each pollutant, yielding unitless proportions that capture the relative contribution of each fleet configuration (Fig. 8). Normalized results reinforce the raw-emission findings by highlighting the disproportionate environmental burden of light trucks. Scenario 1 exhibited the highest normalized contributions across all pollutants, underscoring the structural inefficiency of light vehicles in high-volume biomass logistics. For example, the peak normalized CO2 reached 0.07797, and PMc peaked at 0.08281, representing increases of 148% and 174%, respectively, compared to heavy truck emissions in the same scenario. NO_x followed a similar pattern, with a maximum of 0.0747, 132% higher than the heavy truck counterpart. By contrast, Scenario 2 significantly reduced emission peaks, indicating that mixed fleets can mitigate environmental impacts without full reliance on heavy trucks.

Scenario 3, with mostly medium trucks and one light and one heavy truck, showed that even a single light truck could substantially increase emissions. CO_2 peaked at 0.0664 and PMc at 0.0705, 37–49% higher than medium truck maxima, whereas the heavy truck maintained the lowest emissions (e.g., NO_x 0.00246, SO_2 0.01685), demonstrating the benefits of selective heavy truck deployment. Scenario 4, operated exclusively with heavy trucks, produced stable and comparatively low emissions (CO_2 average 0.0163; PMc range 0.00062–0.03549), illustrating how uniform heavy truck deployment mitigates environmental variability.

In Scenario 5, which uses only medium trucks, emissions were intermediate, higher than those of heavy trucks but substantially lower than those of light trucks. Peak CO_2 and PMc were 0.04835 and 0.04732, 38–42% below the light truck extremes in Scenario 1, providing a balanced environmental outcome for routes unsuitable for heavy vehicles.

Overall, the analysis highlights that truck type, load capacity, and technological level strongly influence environmental performance. Heavy trucks provide the lowest emissions when fully loaded and efficiently routed, while medium trucks offer a practical compromise. Light trucks, when used extensively, lead to the highest emissions. These findings offer robust, policy-relevant insights for sustainable fleet composition and route optimization in BSCs.

4.2. *Emissions cost analysis*

To quantify the environmental cost of biomass transportation emissions, this study applied established social cost values per gram for key pollutants: CO_2 at USD 0.00005/g (~USD 50 per tonne) [66], NO_x at USD 0.012/g, SO_2 at USD 0.015/g [67], and PMc at USD 0.05/g [68]. Fig. 9 presents the emission costs across five scenarios, reflecting variations in truck type, load, and route deployment.

Scenario 1, dominated by light container trucks, exhibited the highest emission costs, with peak values reaching USD 7.46k, nearly 2.5 times higher than the heaviest truck in the same scenario (USD 3.01k). Even the lower-cost light trucks incurred substantial expenses, highlighting inefficiencies due to partial loading and longer travel distances. In contrast, Scenario 2, which combines medium and heavy trucks, produced a more balanced cost profile, demonstrating that strategic deployment of higher-capacity vehicles substantially reduces emission-related expenses. Scenario 3 reinforced this pattern: the inclusion of a single light truck resulted in disproportionately high costs, while heavy trucks consistently maintained low emission costs, confirming their economic and environmental advantage.

Scenario 4, exclusively using heavy trucks, resulted in the lowest and most consistent emission costs (USD 0.06–3.53k, total USD 14.20k), reflecting economies of scale and operational efficiency from

Table 3
Predictive scenario configurations for Woody BSC transportation network.

Supplier-Level type	Type of Transportation Mode for Mass Flow		
	Light	Medium	Heavy
Scenario 1			
S 1-L 1	×	×	✓
S 1-L 2	✓	×	×
S 2-L 1	✓	×	×
S 2-L 2	✓	×	×
S 3-L 1/2	✓	×	×
S 4-L 1	✓	×	×
S 4-L 2	✓	×	×
S 5-L 1	✓	×	×
S 5-L 2	✓	×	×
Scenario 2			
S 1-L 1	×	✓	
S 1-L 2	×	×	✓
S 2-L 1	×	×	✓
S 2-L 2	×	×	✓
S 3-L 1/2	×	×	✓
S 4-L 1	×	✓	×
S 4-L 2	×	×	✓
S 5-L 1	×	×	✓
S 5-L 2	×	×	✓
Scenario 3			
S 1-L 1	✓	×	×
S 1-L 2	×	✓	×
S 2-L 1	×	✓	×
S 2-L 2	×	✓	×
S 3-L 1/2	×	✓	×
S 4-L 1	×	×	✓
S 4-L 2	×	✓	×
S 5-L 1	×	✓	×
S 5-L 2	×	✓	×
Scenario 4			
S 1-L 1	×	×	✓
S 1-L 2	×	×	✓
S 2-L 1	×	×	✓
S 2-L 2	×	×	✓
S 3-L 1/2	×	×	✓
S 4-L 1	×	×	✓
S 4-L 2	×	×	✓
S 5-L 1	×	×	✓
S 5-L 2	×	×	✓
Scenario 5			
S 1-L 1	×	✓	×
S 1-L 2	×	✓	×
S 2-L 1	×	✓	×
S 2-L 2	×	✓	×
S 3-L 1/2	×	✓	×
S 4-L 1	×	✓	×
S 4-L 2	×	✓	×
S 5-L 1	×	✓	×
S 5-L 2	×	✓	×

standardized vehicle deployment. Scenario 5, relying solely on medium trucks, exhibited the widest range of emission costs (USD 0.19–10.74k, total USD 43.21k), indicating heightened financial and environmental risk when medium-capacity fleets are used without optimized routing and load consolidation.

Fig. 10 compares the share of each emission type across suppliers for all scenarios, highlighting the environmental cost distribution within the woody BSC. In Scenario 1, Supplier 5 level 2 recorded the highest total emission cost at USD 7.46 k, driven primarily by CO_2 (USD 6.50k) and PMc (USD 0.35k), indicating either heavy transport loads or extended distances. Similar patterns persisted across scenarios, with Supplier 5 Level 2 and Supplier 1 Level 1 frequently emerging as dominant contributors, while Supplier 2 Level 2 consistently incurred minimal costs, suggesting more efficient routing or proximity advantages. Scenario 3 exhibited elevated costs across nearly all suppliers, reflecting increased reliance on medium and light trucks and less efficient delivery patterns. By contrast, Scenario 4 showed a more controlled and concentrated cost distribution, reinforcing the benefits of high-capacity vehicle standardization.

Across all scenarios (Fig. 11), emission costs were strongly influenced by supplier tier, fleet composition, and routing efficiency. High-volume and long-distance suppliers consistently dominated total emission costs, particularly under scenarios with greater reliance on light or medium trucks. These findings demonstrate that emission cost mitigation in forest BSC depends not only on vehicle technology but also on strategic fleet composition, supplier allocation, and load balancing. Heavy trucks offer the most cost-effective solution when fully utilized, while medium trucks require careful operational optimization to avoid elevated environmental and economic burdens.

4.3. Transportation cost analysis

Fig. 12 compares transportation costs across five scenarios using different truck types, expressed as percentages of the total scenario cost. In Scenario 1, light trucks dominated most routes, contributing the majority of the cost, with the two costliest routes accounting for approximately 47% and 31% of the total, while other light truck entries were minor. Heavy trucks were used only once, representing about 9% of the total. This scenario demonstrates high variability in cost, reflecting inconsistencies in route efficiency.

Scenario 2 featured a mix of medium and heavy trucks, with heavy trucks appearing in most routes and the most expensive heavy truck route contributing around 38% of the total cost. Medium trucks accounted for roughly 27% of the costliest route, and costs varied widely, indicating the importance of strategic routing and load consolidation to avoid inefficiencies.

In Scenario 3, medium trucks represented the majority of routes, with the two most expensive routes contributing approximately 53% of the total cost. Light and heavy trucks were minor contributors, at around 18% and a very low percentage, respectively, suggesting that optimized dispatching could lead to cost savings. Scenario 4 relied exclusively on heavy trucks, which showed high-cost predictability. Standardizing vehicle type and fully utilizing load capacity reduced cost variability and enhanced operational efficiency.

Scenario 5, which relied solely on medium trucks, recorded the highest total transportation cost. Two expensive routes represented approximately 56% of the total, while other routes contributed significantly, indicating that medium trucks can become costly without optimized route planning. Overall, the analysis highlights that truck type and fleet composition strongly influence transportation cost predictability and efficiency. Heavy trucks deliver lower and more consistent costs, while light and medium trucks increase variability and operational risk. Effective routing, load management, and vehicle standardization are critical for achieving cost-efficient biomass logistics.

4.4. Sensitivity analysis of technological development and governance plans

Table 4 outlines the sensitivity analysis design, incorporating diesel, hybrid, and electric trucks; government investment levels (low, moderate, high); fuel price variations; and target years 2030, 2040, and 2050. This framework evaluates environmental (CO_2, NO_x, SO_2, PMc) and economic (transportation cost) performance under varying policy and technological trajectories, with the 2030 diesel–moderate investment scenario serving as the baseline. Results are interpreted in terms of absolute values, percentage changes, and cost–emission trade-offs to capture the combined impact of technology adoption and governance strategies.

The analysis of emissions and transportation costs in the Australian woody BSC from 2030 to 2050 (Fig. 13) reveals clear trends driven by technology choices. Conventional diesel trucks consistently generate the highest CO_2 emissions, with light trucks reaching approximately 31 t/unit/year, medium trucks around 18 t/unit/year, and heavy trucks near 11.6 t/unit/year in 2030, with only marginal decreases by 2050. Hybrid

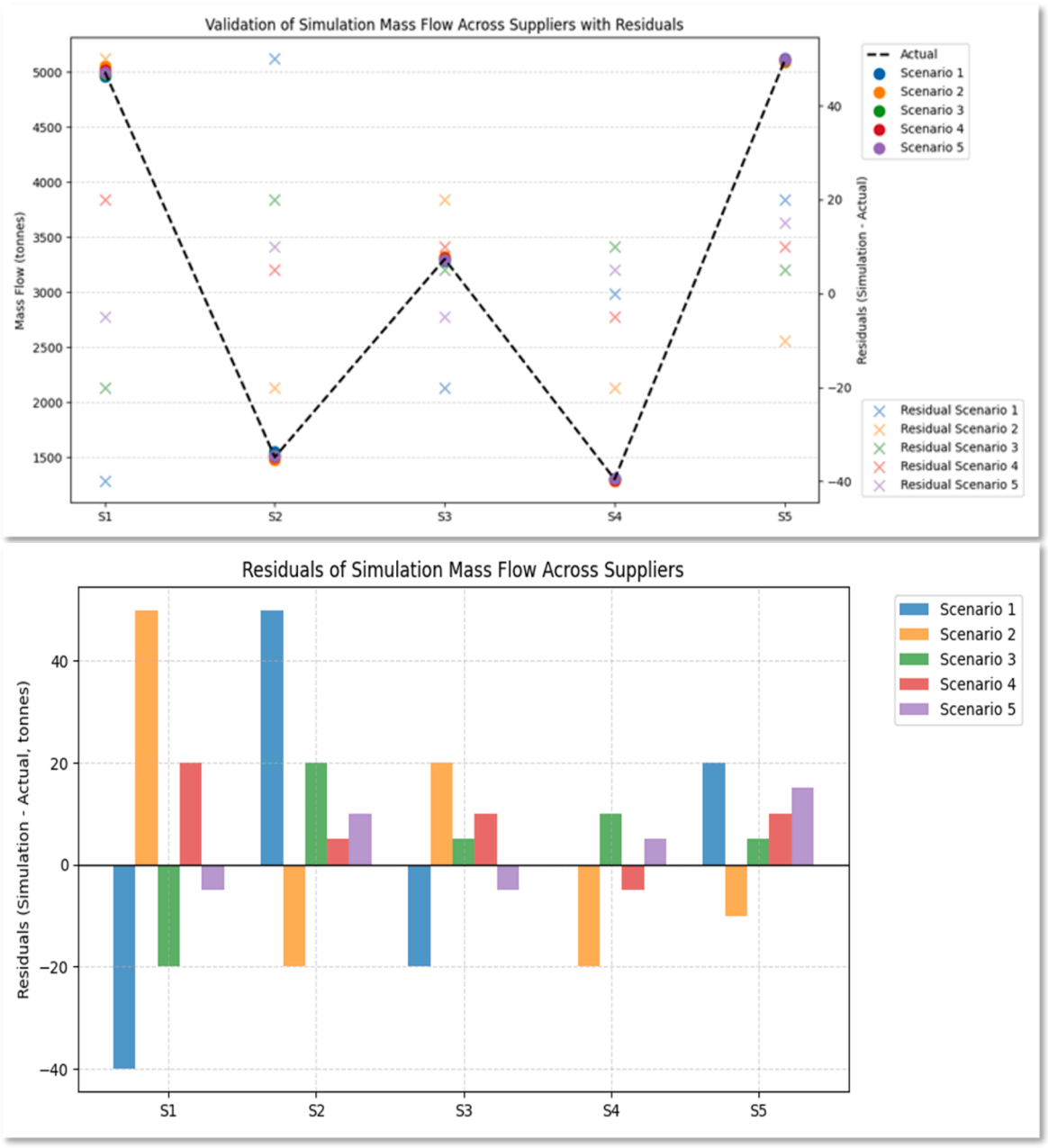


Fig. 6. Validation of Simulation Model.

trucks reduce CO_2 to about 9.8 t/unit/year, while electric trucks achieve the lowest emissions, approximately 2.6 t/unit/year, remaining stable across the period. Comparable patterns are observed for NO_x, SO_2, and PMc, with electric trucks showing the most substantial reductions, particularly in particulate matter, which drops below 2 t/unit/year by 2050.

Transportation costs exhibit an inverse trend to emissions. Light diesel trucks incur the highest costs (~USD 22k), whereas heavy diesel trucks remain lower (USD 10–11k). Hybrid and electric trucks maintain moderate costs (USD 11–12k), with electric vehicles initially slightly more expensive than medium diesel trucks but stable over time due to lower fuel price sensitivity. Efficiency gains, lower fuel sensitivity, and potential policy incentives make electric trucks increasingly viable over time. Policy investment scenarios strongly influence outcomes. Low investment slows hybrid and electric adoption, maintaining high emissions, whereas moderate investment enables significant hybrid uptake by 2040. High investment accelerates electrification, achieving near-maximal emission reductions by 2050.

Operational uncertainties, such as fuel price volatility, disproportionately affect diesel trucks, potentially increasing costs by 5–10%, whereas electric trucks provide predictable expenses alongside environmental benefits.

Fig. 14 illustrates the cumulative emission reduction potential of various truck types by 2050 across four key pollutants (CO_2, NO_x, SO_2, and PMc). The 3D surface plot illustrates the comparative performance of each transportation mode in relation to the baseline light-duty truck. As shown, electric trucks achieve the highest overall reduction, with an 88.5% decrease in cumulative emissions compared to the baseline, followed by hybrid trucks (69.2%), heavy trucks (52.3%), and medium trucks (38.4%). The pronounced gradient on the surface indicates substantial emission mitigation associated with the shift from conventional diesel toward hybrid and electric technologies. These findings highlight the transformative potential of electrification and hybridization in reducing the environmental impact of freight transport. Moreover, they

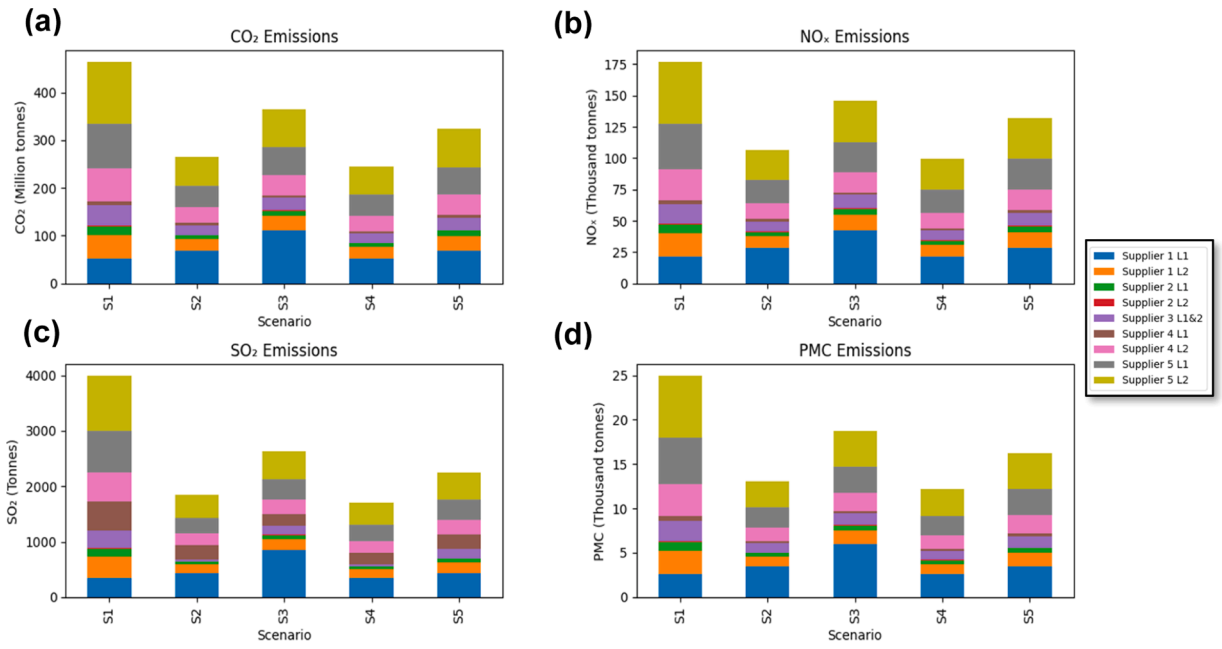


Fig. 7. Environmental emissions analysis of suppliers in tier 1 and tier 2.

reinforce the importance of integrating clean truck technologies within sustainable biomass and logistics systems to support Australia's net-zero emission targets by 2050. Additional breakdowns for each truck category and pollutant are provided in Appendix (Figs. A2(a–f)) for detailed comparison.

Fig. 15 highlights cost gradients: heavy trucks remain most cost-effective (ranging from AU$0.98–1.49 million), hybrid trucks incur a modest 15–20% premium, and electric trucks incur a slightly higher cost due to capital and payload constraints.

Between 2030 and 2050, transportation costs for hybrid and electric trucks increase by approximately 20–25%, consistent with projected growth in biomass flows, whereas costs for heavy trucks rise by only ~10%, highlighting their operational efficiency. Overall, the sensitivity analysis underscores the transformative potential of electrification and hybridization for sustainable biomass logistics. Integrating clean truck technologies, supported by strategic policy investment, can reduce environmental impacts substantially while maintaining economic competitiveness. These insights provide a roadmap for low-carbon SC planning in Australia and similar forestry-based biomass systems globally.

4.5. Discussion

The results demonstrate that truck type, route alignment, and fleet deployment are key determinants of emissions and transportation costs in the woody BSC. Conventional diesel trucks maintain relatively high CO_2, NO_x, SO_2, and PMc emissions throughout 2030–2050, with light trucks producing the highest emissions (~31 t CO_2 per unit in 2030) and medium/heavy trucks achieving moderate reductions. Hybrid trucks provide intermediate mitigation (~9.8 t CO_2), whereas electric trucks achieve the lowest emissions (~2.6 t CO_2), consistent across all time horizons. Similar patterns are observed for NO_x, SO_2, and PMc, with electric vehicles reducing particulate matter below 2 t/unit by 2050.

Economic assessments reveal an inverse trend: light diesel trucks incur the highest transportation costs, heavy diesel trucks remain cost-efficient, and hybrid/electric trucks exhibit moderate costs, reflecting capital and battery-related constraints offset by lower fuel sensitivity and potential policy incentives. These findings highlight the trade-off between cost efficiency and environmental performance, with electric trucks providing substantial sustainability gains at manageable additional costs.

Scenario analyses and sensitivity tests further underscore the importance of government investment and operational uncertainty. Low investment delays hybrid/electric adoption, maintaining higher emissions, while high investment, including subsidies, charging infrastructure, and policy incentives, accelerates electrification, achieving near-maximal emissions reductions by 2050. Fluctuating fuel prices disproportionately affect diesel trucks (5–10% cost increase), whereas electric trucks provide predictable operational costs, emphasizing their resilience under uncertainty.

Supplier-level analysis reveals key emission hotspots, with Supplier 5 Level 2 consistently generating the highest emissions due to long-haul distances and suboptimal vehicle allocation. While light trucks offer operational flexibility, they are cost-inefficient for high-volume or long-distance deliveries. These findings emphasize the critical role of strategic vehicle-load alignment, route optimization, and supplier management in minimizing both emissions and transportation costs.

Prior research in wood and biomass transport has substantially improved understanding of logistics efficiency, coordination, and environmental performance. Early simulation studies demonstrated that improved information visibility and adaptive dispatching significantly enhance system utilization, yet these analyses largely emphasized short-term operational efficiency rather than sustainability transitions [47]. The DES models of biomass supply chains have evaluated harvesting, transport, storage, and demand-driven logistics using cost and GHG metrics. While these studies advanced cost–emission trade-off analysis, they typically relied on fixed structural assumptions and focused on tactical optimization rather than adaptive fleet transformation under evolving policy and technology conditions [48]. Vehicle configuration research further showed that higher-capacity truck–trailer systems can reduce unit transport cost and fuel intensity. However, such analyses generally compare static configurations and do not embed fleet choices within dynamic, system-wide transition frameworks [49].

More recently, integrated GIS and agent-based modelling approaches have incorporated spatial heterogeneity and multimodal decisions, demonstrating the environmental benefits of increased rail share and optimized routing. Despite this progress, these models primarily evaluate comparative scenarios and rarely address long-term electrification pathways or decentralized decision-making under investment and fuel price uncertainty [39].

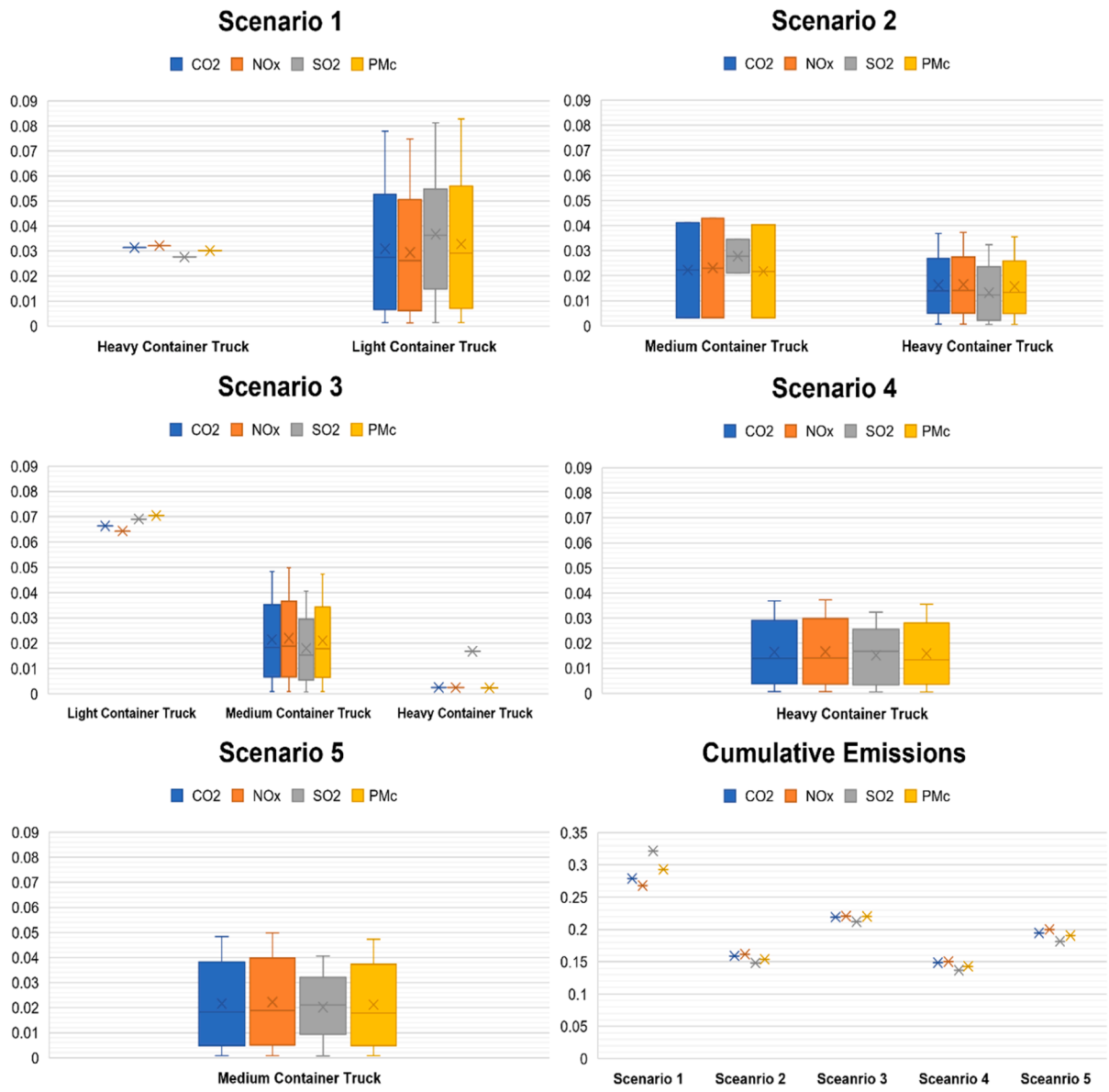


Fig. 8. Comparative analysis of transportation emissions for each scenario.

Parallel to forest transport research, a growing body of literature examines heavy-duty truck electrification using life cycle assessment (LCA) and total cost of ownership (TCO) frameworks. Comparative analyses in China demonstrate that battery electric, plug-in hybrid, hydrogen fuel cell, and catenary trucks exhibit substantially different life-cycle emissions depending on electricity and hydrogen pathways, with green energy sources playing a decisive role [69]. Urban freight simulation studies show that battery electric trucks significantly reduce air toxics, while hybrid configurations often provide better short-term cost performance; large-scale deployment requires infrastructure and policy support [70]. European comparative assessments indicate that battery-electric trucks—particularly under high-power charging or electric road systems—achieve the lowest life-cycle emissions and costs, whereas fuel-cell trucks remain advantageous for long-haul operations [71].

Similarly, U.S. technoeconomic studies reveal that although battery-electric trucks currently involve higher private costs, they generate substantially lower external climate and health damages, leading to positive societal benefits under supportive policies [72]. These studies clarify technology-level cost–emission trade-offs and infrastructure sensitivity. However, they largely evaluate electrification at vehicle or national system levels, without embedding these technologies within spatially explicit, supply chain–oriented forest logistics models.

Building on prior wood transport simulation research and recent heavy-duty truck electrification studies, this work advances forest and wood SC analytics through the development of an integrated GIS–agent-based DT framework that embeds diesel, hybrid, and electric trucks within a spatially explicit biomass logistics system. Unlike discrete-event models that primarily optimize utilization, scheduling, or modal mix, and unlike aggregate LCA–TCO analyses that evaluate vehicle technologies at national or fleet-wide levels, the proposed approach simultaneously captures decentralized supplier behaviour, adaptive routing decisions, long-term transition pathways (2025–2050), and policy and fuel-price uncertainty within a unified modelling environment. By enabling concurrent evaluation of operational cost, particulate and gaseous emissions, routing structure, and fleet composition, the framework shifts the analytical focus from static vehicle comparison or short-term operational efficiency toward predictive sustainability transition modelling.

Furthermore, it conceptualizes fleet restructuring as a strategic lever

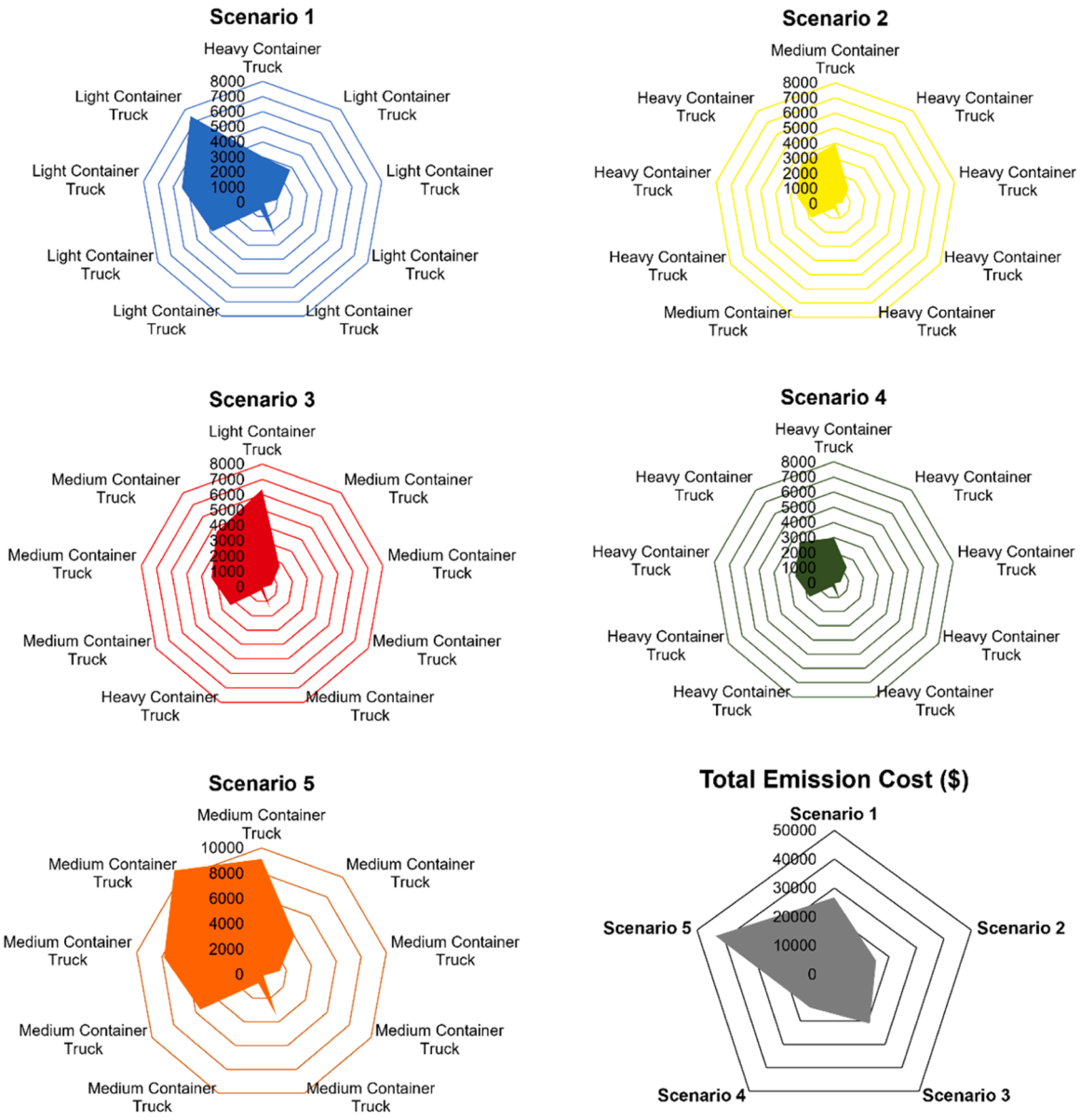


Fig. 9. Emission cost analysis by transportation type of each scenario.

aligned with government incentives and net-zero objectives rather than merely an operational adjustment. Through embedding environmental externalities directly into operational analytics, the DT architecture transforms simulation-based forest transport modelling into a strategic, sustainability-oriented decision-support system, thereby clarifying the manuscript's incremental contribution and positioning it at the intersection of forest SC management, transport electrification, and supply chain analytics.

4.5.1. Practical, managerial, and policy implications

The study provides actionable insights for SC managers, transport operators, and policymakers seeking low-emission, cost-effective logistics. Key recommendations include:

- Fleet composition: Prioritize heavy, hybrid, and electric trucks for long-haul, high-volume operations; light trucks may serve short-distance, low-volume deliveries. Align vehicle type with supplier locations, load capacities, and route lengths to optimize emissions and costs.
- Digital twin and ABM integration: Use real-time simulation to dynamically assign vehicles, optimize routes, and forecast emissions under varied operational scenarios. This enables predictive logistics management and improves decision-making.
- Policy incentives: Targeted government measures, such as subsidies for hybrid/electric trucks, investment in charging infrastructure, green logistics certifications, and strategic supplier zoning, accelerate low-emission adoption while preserving economic efficiency.
- Emission management: Pair high-emission suppliers with cleaner vehicles or implement carbon-offset mechanisms to achieve SC-wide emission reduction targets.
- Data-driven planning: Integrate transport emissions databases within ABM–GIS frameworks for benchmarking, compliance reporting, and long-term infrastructure planning.

Beyond the biomass context, this study contributes to forest and wood SC research by demonstrating how simulation-based operational models can evolve into strategic sustainability decision-support systems. The proposed framework complements and extends prior wood transport simulation studies by embedding environmental metrics, policy

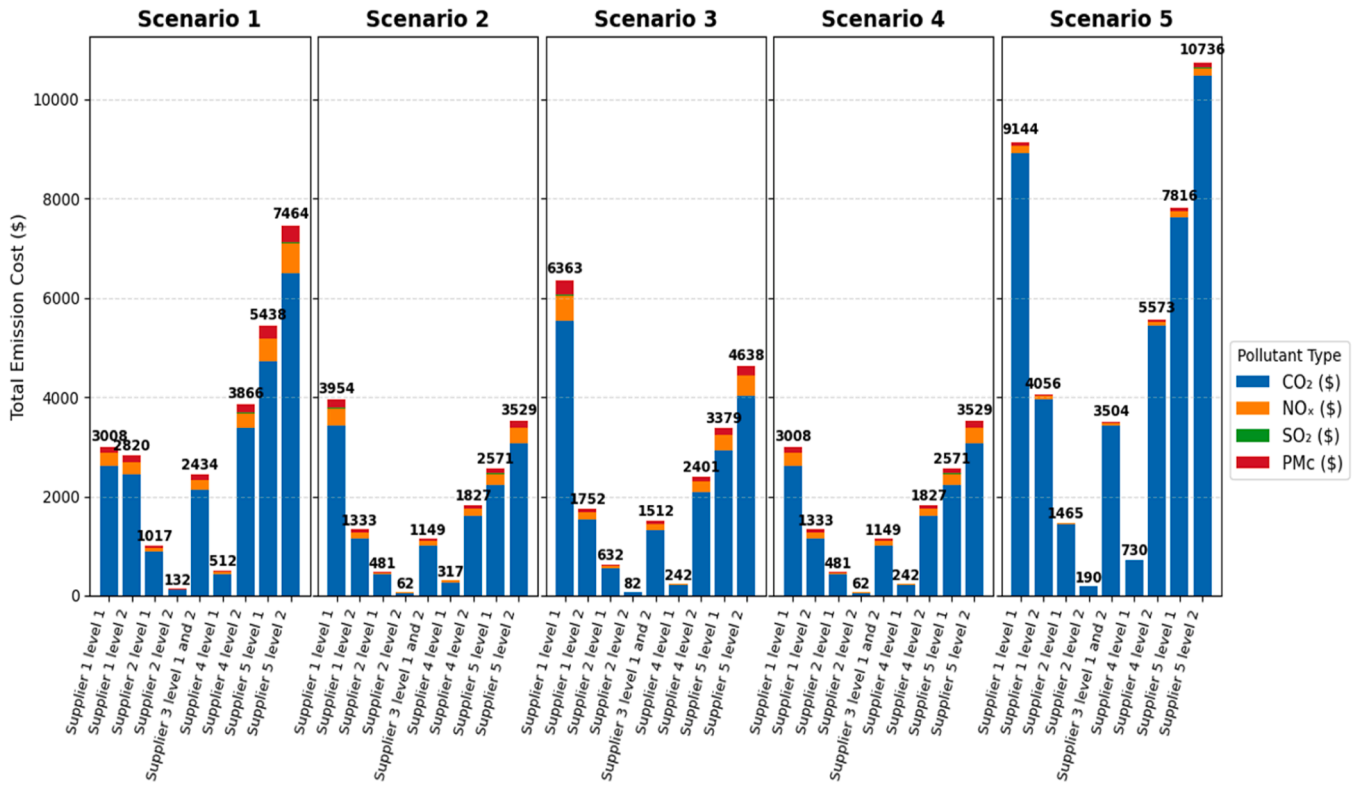


Fig. 10. Emissions cost analysis for each supplier of scenarios.

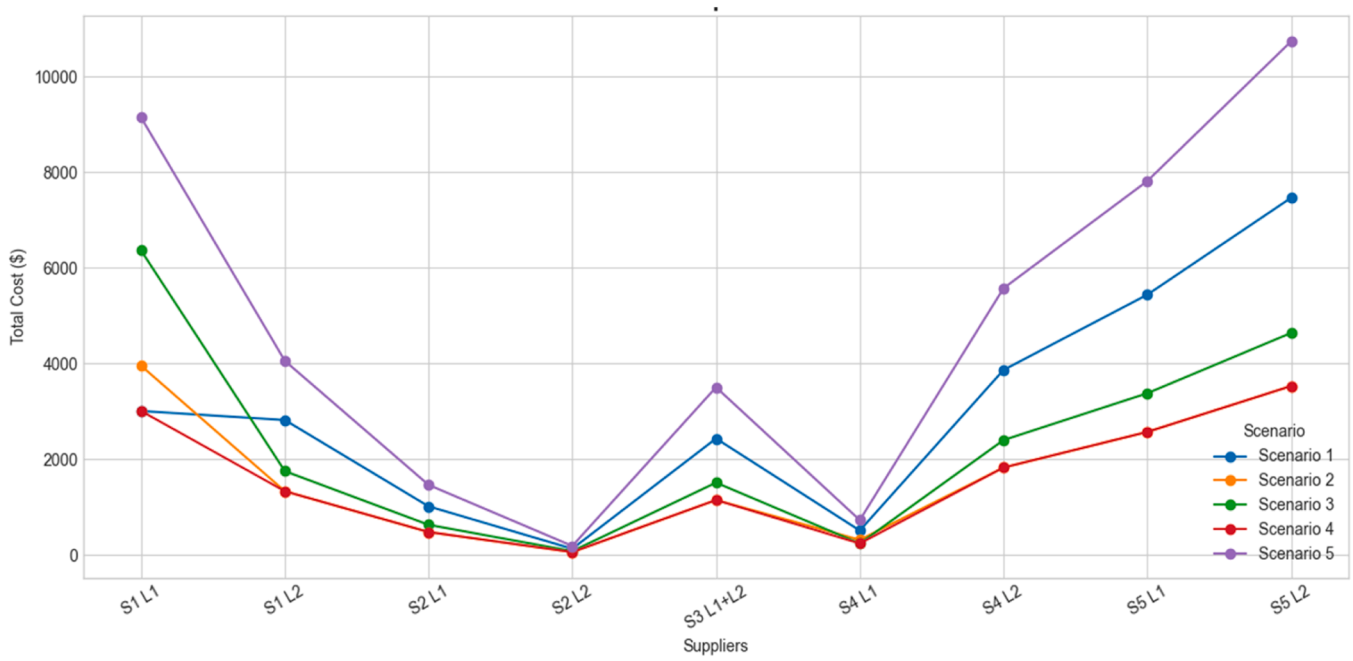


Fig. 11. Total emissions cost comparison across scenarios.

dynamics, and technology transition pathways directly into operational analytics.

Although this study focuses on the biomass sector in Victoria, the principles are transferable to other industries (food, healthcare, general freight) and geographic contexts facing similar mobility transitions. Leveraging electrification, predictive analytics, and DT systems enables policymakers and managers globally to build sustainable, resilient, and cost-efficient logistics networks aligned with future low-carbon objectives. This study relies on published emission factors, stakeholder reports, and historical data; no real-time stakeholder meetings were conducted. Scenario outcomes may vary under different supplier behaviors, traffic conditions, or regional policies. Nevertheless, the ABM–GIS framework is flexible and can incorporate updated empirical data, alternative fleet compositions, and policy scenarios for broader applicability.

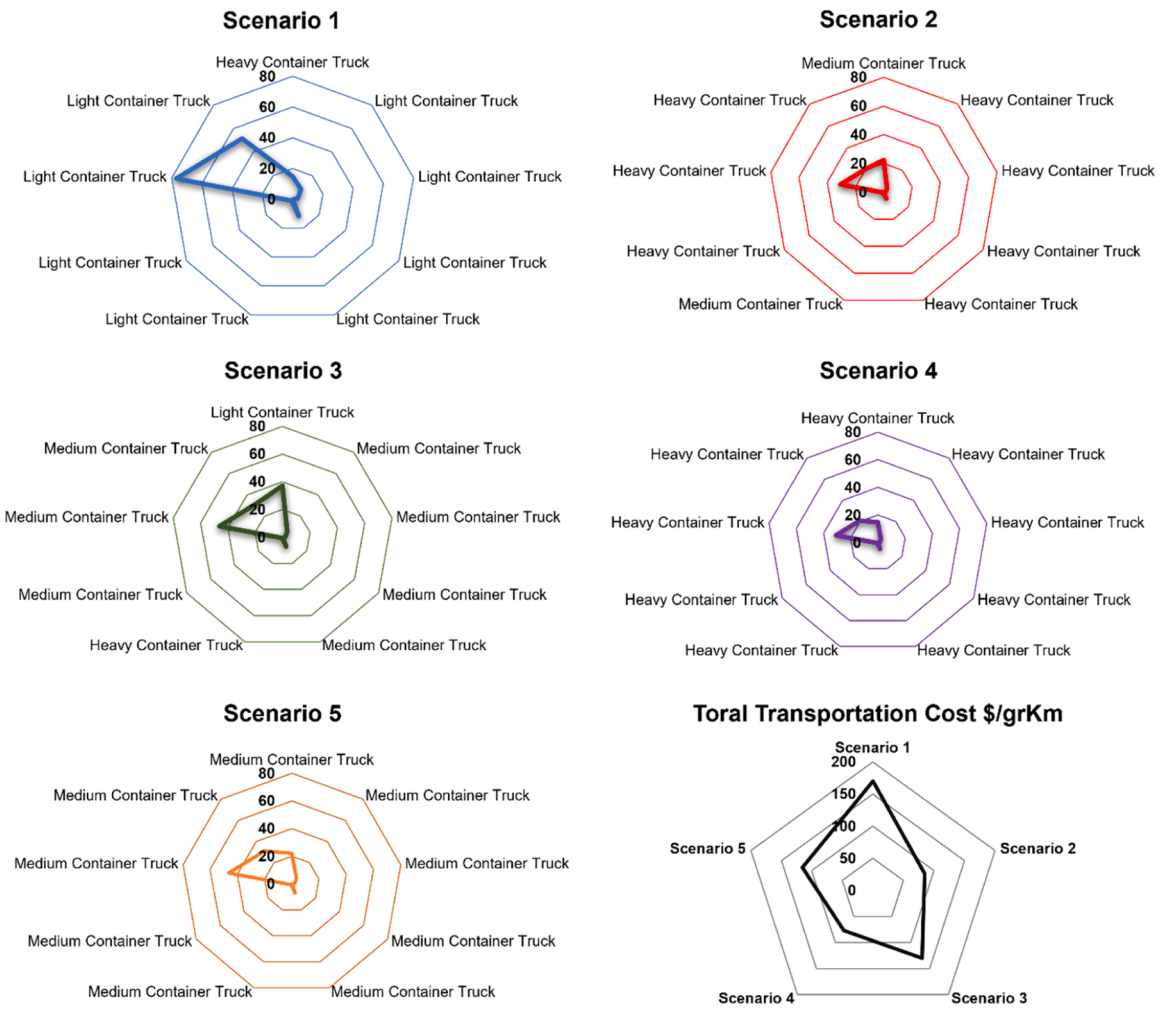


Fig. 12. Transportation cost analysis of each scenario based on the type of trucks.

Table 4
Sensitivity analysis design for sustainable mobility assessment.

Parameter	Categories / Values Considered
Transportation Mode	Current Diesel Truck, Hybrid Truck, Electric Truck
Target Year	2030, 2040, 2050
Government Investment Level	Low, Moderate, High
Environmental Indicators	CO_2, NO_x, SO_2, PMc
Economic Indicator	Transportation Cost (AUD)
Baseline Scenario	Current Diesel Truck – 2025 – Current Investment
Comparison Metrics	Absolute values, % change vs. baseline, cost-emission trade-offs

5. Conclusion and future works

This study set out to assess how different truck technologies—diesel, hybrid, and electric—combined with alternative supplier–tier allocations, affect the sustainability of forest BSCs in Victoria, Australia, from 2025 to 2050. To achieve this, a digital twin–ready framework was developed by integrating ABM and GIS, enabling detailed simulation of transport routes, truck class allocations, emissions, and transport costs across light, medium, and heavy vehicles.

Validation against empirical supply volumes confirmed the model's robustness, with deviations ranging from 0.5 to 3%. By assessing both environmental indicators (CO_2, NO_x, SO_2, and PMc) and economic performance (transportation cost) across multiple scenarios, truck types, and suppliers, the analysis demonstrates that heavy trucks consistently achieve the lowest emissions and most stable transportation costs, while medium and light trucks exhibit higher variability and elevated emissions, particularly when route alignment is inefficient.

Hybrid and electric trucks provide substantial emission reductions up to 80–90% for CO_2 and PMc, at competitive costs, highlighting their pivotal role in decarbonizing the BSC. Government investment further strengthens these outcomes by accelerating technology adoption, mitigating fuel-price sensitivity, and improving the overall cost–emission balance. These findings advance digital twin-enabled SC management by illustrating how predictive, scenario-based simulations can support sustainable, cost-efficient logistics planning.

Hybrid trucks show intermediate benefits, reducing emissions by approximately 50–81% depending on vehicle class, which is broadly in line with findings reported by Kumar and Paul (2023) for biomass and freight transport systems. Unlike prior LCAs and transport studies that often consider emissions or costs in isolation, our framework simultaneously evaluates environmental and economic trade-offs, revealing that heavy electric trucks can achieve substantial emission reductions with only a moderate cost increase (~20%) by 2050. These comparisons demonstrate that our study not only corroborates the trends observed in earlier work but also provides a more comprehensive and spatially explicit assessment, offering actionable insights for fleet electrification, supplier strategies, and policy planning in sustainable biomass logistics.

This study has several limitations that should be acknowledged. First, the DT framework was applied in a simulation-ready mode without full real-time data integration, meaning that the dynamic responsiveness of a live DT could not be demonstrated. Second, key

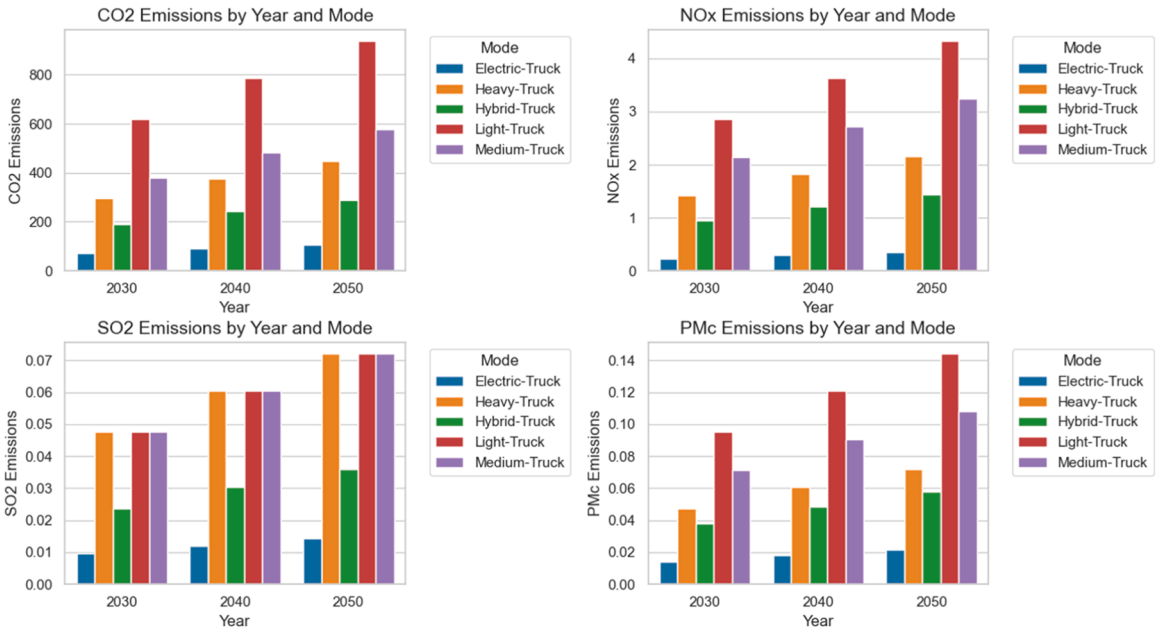


Fig. 13. Environmental Assessment of Mobility vs. Technology and Plans.

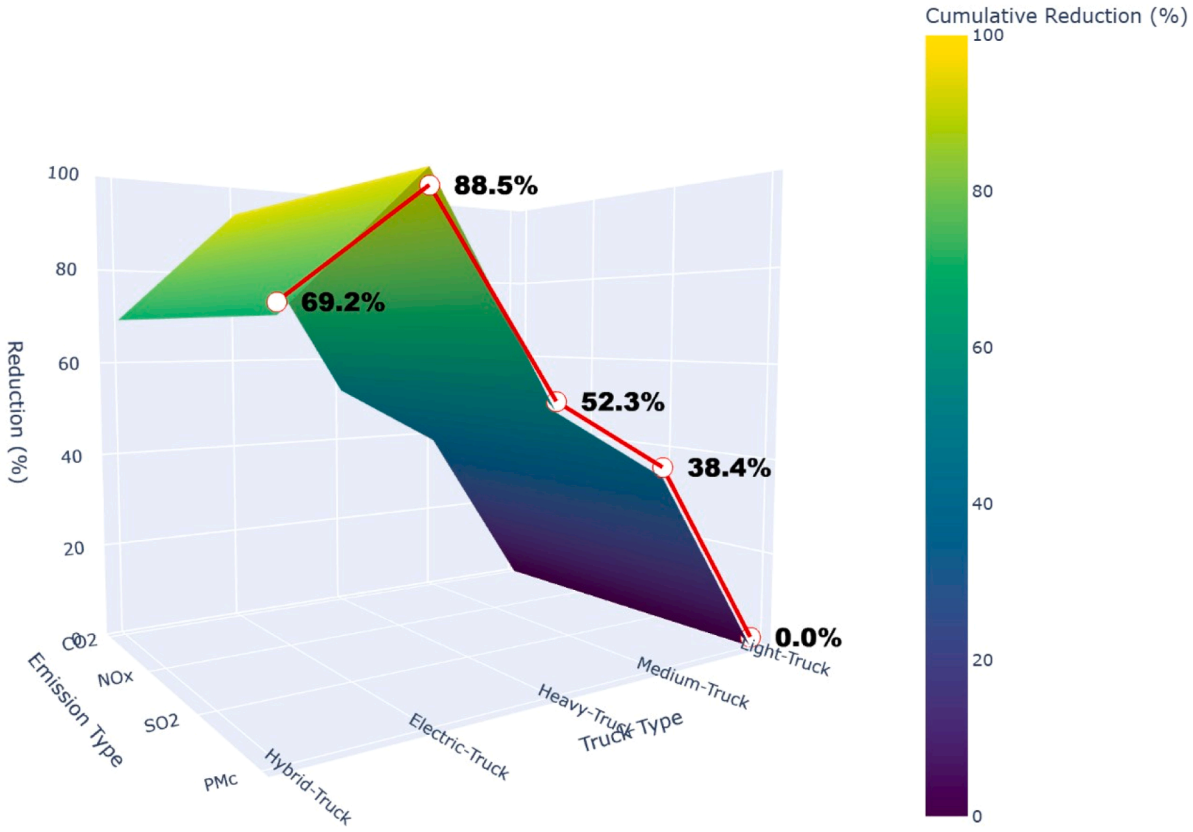


Fig. 14. Comparative cumulative emission reduction of truck types by 2050 across four key pollutants.

assumptions on truck energy consumption, grid decarbonisation, and fuel price trajectories were based on projections that may vary significantly under future policy or technological developments. Third, operational factors such as charging infrastructure availability, vehicle range constraints, and battery replacement costs were simplified and not explicitly modeled, which may understate practical adoption barriers for electric trucks. Fourth, the analysis focused on transport-related emissions and costs, without including upstream impacts such as vehicle manufacturing, battery SCs, or end-of-life considerations. Finally, the model was validated primarily on supply volumes, while additional validation against fuel consumption, route-level emissions, and operational costs would further strengthen robustness.

Future studies should address these gaps by integrating real-time logistics data, expanding the scope of life-cycle impacts, and

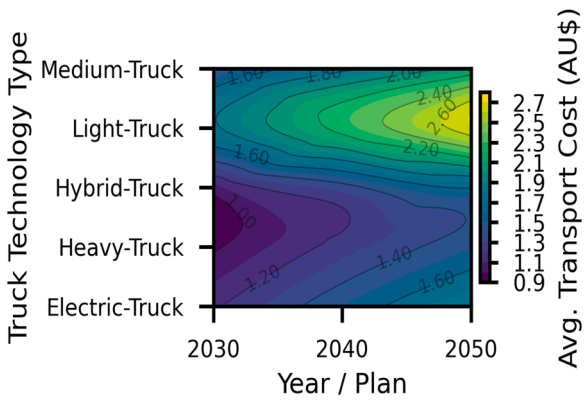


Fig. 15. Average Transportation Cost (Million AU$) vs. Type of Truck (Based on Technology), and Target Year.

embedding optimization modules to support decision-making under uncertainty. Future research should integrate real-time variables such as weather, road accessibility, and traffic congestion to enhance predictive accuracy, evaluate emerging vehicle technologies, including hydrogen and autonomous trucks, and explore multimodal transport strategies. Additionally, simulating dynamic fuel pricing, carbon taxation, and carbon credit mechanisms can guide policy design, while incorporating IoT sensor data and machine-learning algorithms would enable adaptive route optimization and decision support. Expanding the framework across different geographies and SC sectors, such as food and healthcare, will further validate its scalability and generalizability. Finally, the development of an interactive DT dashboard with intuitive KPIs can provide actionable insights for policymakers, industry stakeholders, and logistics managers. Collectively, this study establishes a robust foundation for evidence-based decision-making in sustainable biomass logistics, delineating a clear pathway toward low-carbon, cost-efficient transportation systems by 2050.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.The authors declare no competing financial interests.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.sca.2026.100203.

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